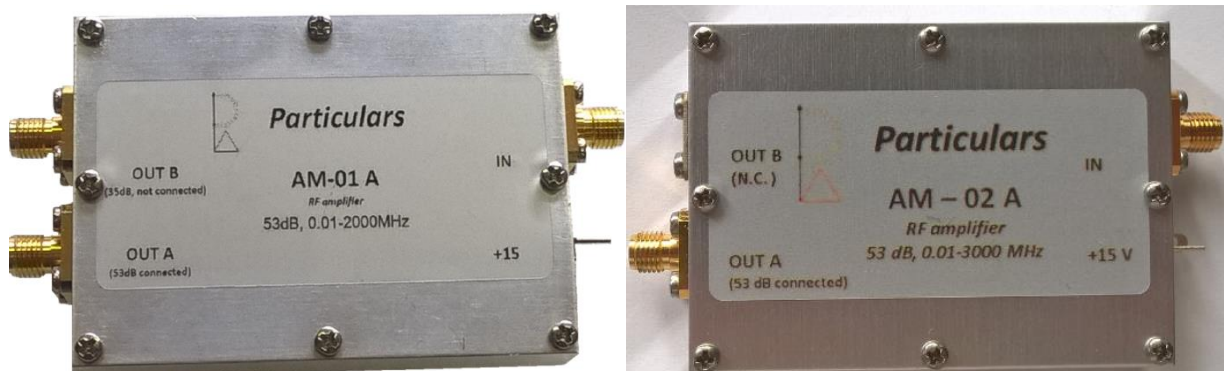


Particulars Wide Band Current Amplifiers

Introduction



Particulars wide band width current amplifiers are designed for the use in transient current systems. Regarding of the intended use they have nominal amplifications of 12, 35 and 53 dB. The 12 and 33 dB are used in Compact-TCT setups, while 53 dB is intended for use in Scanning-TCT systems where focus light results in high density of free carriers. In order to minimize the effect of this to device performance the concentration should be kept low which requires on the other hand high amplification.

Models

AM-01 A,B – no longer produced (Q1-2017)

AM-02 A,B – current version of the amplifiers

Connections

IN – input of the pulse (50 Ω)

+15 V – power of the amplifier. Note that 15 V is the max. voltage to be applied (smaller gain with marginal degradation in speed for lower bias). The pin should be soldered to the power-supply while the chassis should be grounded. The bias voltage for AM-02A should be 6-13 V.

Out A – output of the amplifier (50 Ω) for the 53 dB version of the amplifier

Out B – output of the amplifier (50 Ω) for the 12, 33 dB version of the amplifier

Response function of the amplifier – S parameters

The response of the amplifier as measured with the network analyser are as follows.



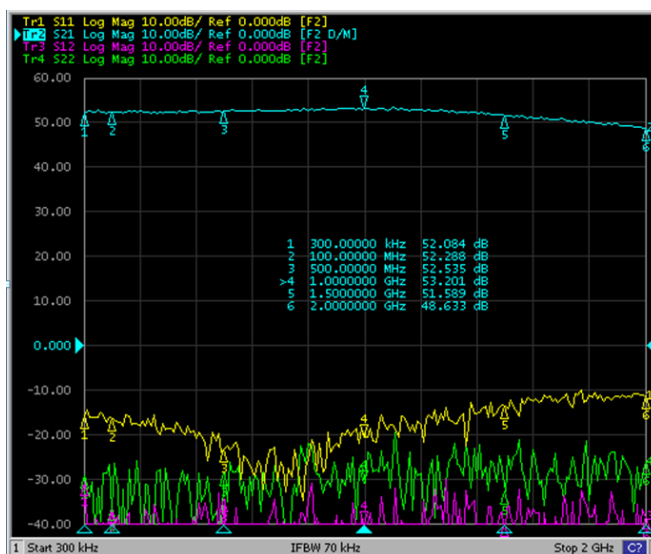
a1 - input signal

b1 - reflected signal

b2 - output signal

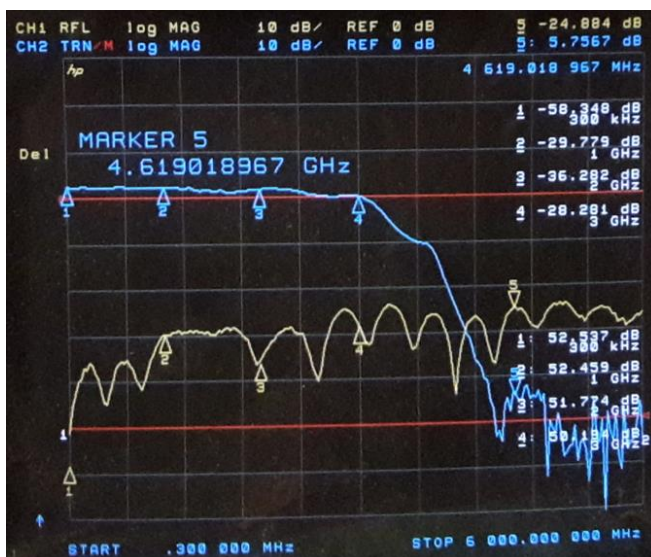
$$\begin{pmatrix} b_1 \\ b_2 \end{pmatrix} = \begin{pmatrix} S_{11} & S_{12} \\ S_{21} & S_{22} \end{pmatrix} \begin{pmatrix} a_1 \\ a_2 \end{pmatrix}$$

S21 - Gain of the amplifier



Network analyser measurements

AM-01A



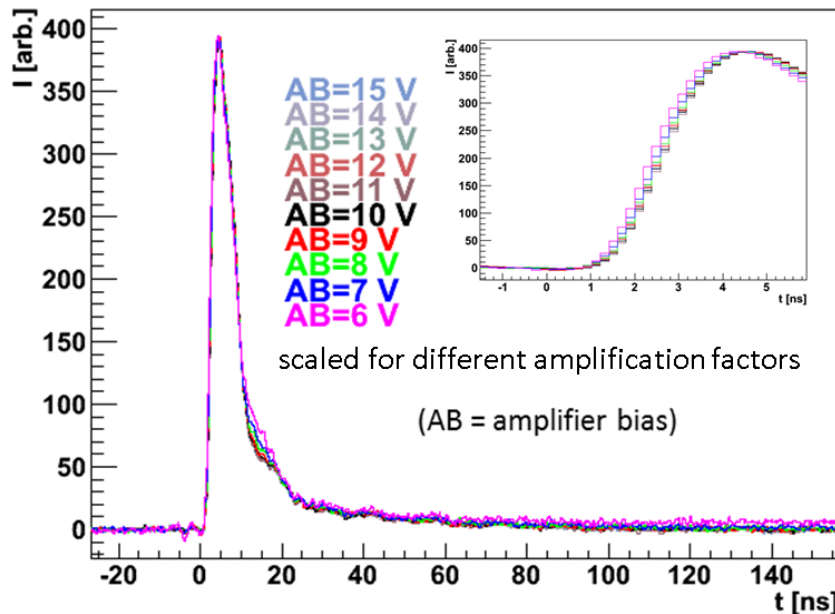
Network analyser measurements

AM-02A → S21 in blue

gain values at different frequencies
are in the bottom of the plot

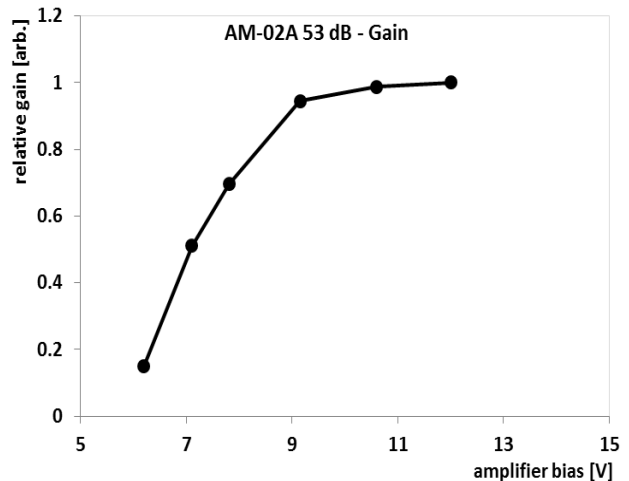
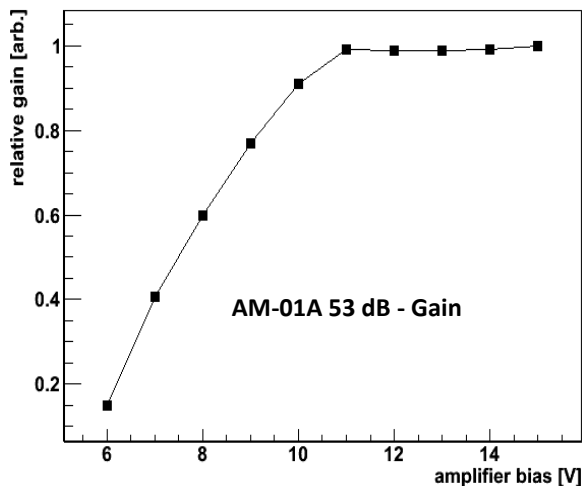


Response – GAIN of the amplifier at different bias voltages (53 dB)



TCT signals in 300 μm fully depleted silicon n-type (float zone) sample with 1064 nm light pulses

The amplification of the amplifier is given at nominal amplifier bias voltage. However the gain can be reduced by lowering the amplifier bias voltage. The response of the 53 dB amplifier with a test sample is shown in the figures below for AM-01A and AM-02A. A reduction of factor 10 can be achieved by reducing the amplifier bias voltage to the lowest stable operational voltage of 6 V (see below figure). No significant impact of amplifier bias voltage on bandwidth has been observed. Nominal operational voltage of AM-02A is around 12 V.





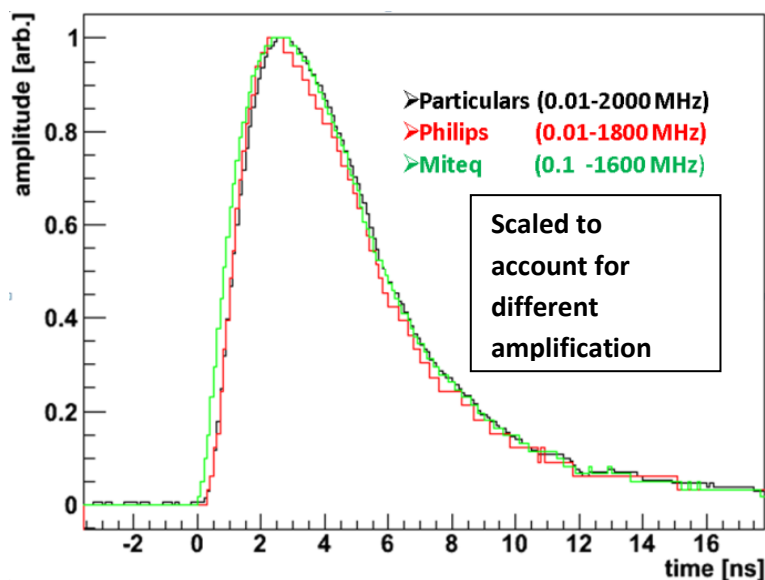
Saturation voltages

Saturation voltages of the amplifier depend on the signal polarity due to asymmetric clipping of the components.

Type of amplifier	Saturation Voltage (neg. signal)	Saturation voltage (pos. signal)
35 dB amplifier	~-1900 mV (AM-01A)	~850 mV (AM-01A)
53 dB	~-1800 mV (AM-01A)	~900 mV (AM-01A)

Comparison to other vendors

No significant different in terms of bandwidth has been observed with wide band width amplifiers from other vendors.



Amplifiers have been tested with infra-red laser (1064 nm) in a fully depleted 300 mm thick n-p silicon sensor (diode).



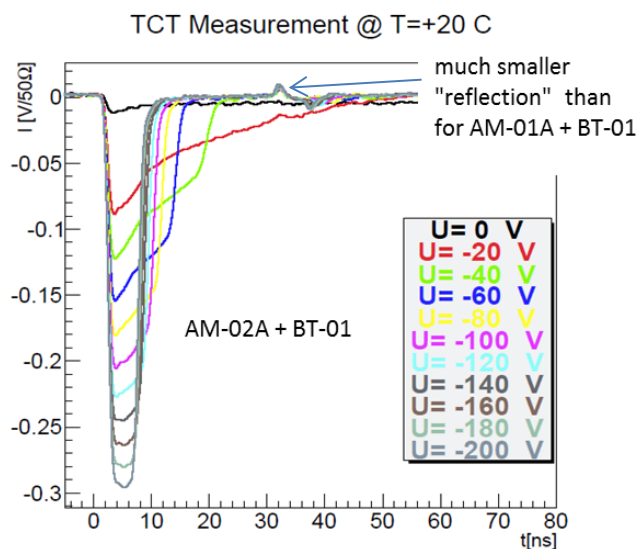
Specifications

model AM-01 A

bias voltage	6-15 V
Amplification at nominal voltage	35 dB, 53 dB
frequency range	0.01-2000 MHz
input impedance	~50 Ω
output impedance	~50 Ω
current	140 mA
Physical measures [cm] (LxWxD)	7x5x1.5

model AM-02 A

bias voltage	6-13 V
Amplification at nominal voltage	35 dB, 53 dB
frequency range	0.01-3000 MHz
input impedance	~50 Ω
output impedance	~50 Ω
current	150-200 mA
Physical measures [cm] (LxWxD)	7x5x1.5



AM-02A has improved impedance matching with BT-01 (Bias-Tee) which mitigates the reflections. The figure shows induced currents for p-n silicon diode of 300 mm ($V_{fd} \sim -20V$) illuminated from the front (electron injection). Note a small distortion of the signal at around 30 ns.